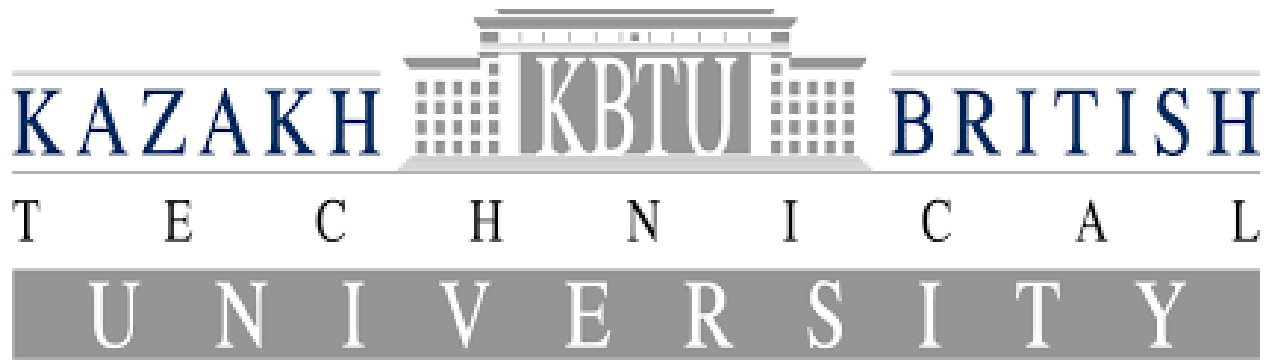




School of Information Technology and Engineering
Security of Operating System

Custom SELinux Policy for the SWIFT Backend Service
“Secure SWIFT Logical Terminal Infrastructure”

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1. Objective

The goal of this task is to design and implement a **custom SELinux Type Enforcement (TE) policy** on **Fedora Linux** for the swiftapi backend service.

Instead of relying on default permissive domains (e.g., `unconfined_service_t`), a **strict least-privilege model** is enforced where:

- Only required files are accessible
- Only a dedicated port is usable
- Only trusted executables can run
- All other actions are denied and logged (AVC)

2. Environment Setup (Fedora)

2.1 Install Required SELinux Tools

```
sudo dnf install selinux-policy-devel checkpolicy policycoreutils-python-utils -y
```

```
valid_tft forever preferred_tft forever
swift-terminal@fedora:/opt/swift-backend$ sudo dnf install selinux-policy-devel checkpolicy policycoreutils-python-utils -y
[sudo] password for swift-terminal:
Updating and loading repositories:
Repositories loaded.
Package "checkpolicy-3.9-2.fc43.x86_64" is already installed.
Package "policycoreutils-python-utils-3.9-5.fc43.noarch" is already installed.

Package Arch Version Repository Size
Upgrading:
selinux-policy noarch 43.7-1.fc43 updates 32.0 KiB
replacing selinux-policy noarch 42.12-1.fc43 0e44b8838a064239 31.7 KiB
selinux-policy-targeted noarch 43.7-1.fc43 updates 18.6 MiB
replacing selinux-policy-targeted noarch 42.12-1.fc43 0e44b8838a064239 18.6 MiB
Installing:
selinux-policy-devel noarch 43.7-1.fc43 updates 23.1 MiB
Installing dependencies:
m4 x86_64 1.4.21-1.fc43 updates 877.2 KiB
policycoreutils-devel x86_64 3.9-5.fc43 fedora 325.5 KiB

Transaction Summary:
Installing: 3 packages
Upgrading: 2 packages
Replacing: 2 packages

Total size of inbound packages is 9 MiB. Need to download 9 MiB.
After this operation, 24 MiB extra will be used (install 43 MiB, remove 19 MiB).
[1/5] policycoreutils-devel-0:3.9-5.fc43.x86_64 100% | 361.4 KiB/s | 139.9 KiB | 00m00s
[2/5] selinux-policy-0:43.7-1.fc43.noarch 100% | 36.5 KiB/s | 73.2 KiB | 00m02s
[3/5] m4-0:1.4.21-1.fc43.x86_64 100% | 109.7 KiB/s | 343.9 KiB | 00m03s
[4/5] selinux-policy-devel-0:43.7-1.fc43.noarch 100% | 380.3 KiB/s | 1.4 MiB | 00m04s
[5/5] selinux-policy-targeted-0:43.7-1.fc43.noarch 100% | 2.1 MiB/s | 6.8 MiB | 00m03s
-----
[5/5] Total 100% | 818.1 KiB/s | 8.8 MiB | 00m11s
Running transaction
[1/9] Verify package files 100% | 41.0 B/s | 5.0 B | 00m00s
[2/9] Prepare transaction 100% | 16.0 B/s | 7.0 B | 00m00s
[3/9] Upgrading selinux-policy-0:43.7-1.fc43.noarch 100% | 3.7 KiB/s | 33.6 KiB | 00m09s
```

These tools provide:

- Policy compilation (checkmodule)
- Packaging (semodule_package)
- SELinux management utilities (semanage, restorecon)

3. Policy Design Overview

3.1 Custom Types

TYPE	PURPOSE
SWIFTAPI_T	Process domain
SWIFTAPI_EXEC_T	Executable
SWIFTAPI_CFG_T	Configuration
SWIFTAPI_DATA_T	SWIFT data
SWIFTAPI_LOG_T	Logs
SWIFTAPI_TMP_T	Temporary files
SWIFTAPI_PORT_T	TCP port 8000

3.2 Security Model

The policy enforces:

- **Read-only access** to config files
- **Full control** over SWIFT data directory
- **Append-only logging**
- **Exclusive port binding**
- **No privilege escalation**

4. Implementation

4.1 Create Policy Directory

```
mkdir -p /home/swift-terminal/selinux-policy/swiftapi  
cd /home/swift-terminal/selinux-policy/swiftapi
```

```
swift-terminal@fedora:~/home$ mkdir -p /home/swift-terminal/selinux-policy/swiftapi
cd /home/swift-terminal/selinux-policy/swiftapi
swift-terminal@fedora:~/selinux-policy/swiftapi$ pwd
/home/swift-terminal/selinux-policy/swiftapi
```

4.2 Create Policy Files

(1) Type Enforcement File

nano swiftapi.te

```

GNU nano 8.5 swiftapi.te
policy_module(swiftapi, 1.0)

require {
    type init_t;
    type bin_t;
    type lib_t;
    type cert_t;
    type node_t;
    type http_port_t;
    type setroubleshootd_t;
    # Required for the port definition to work
    attribute port_type;
    class file { getattr read open getattr execute execute_no_trans map };
    class dir { getattr read search open getattr };
    class tcp_socket { create ioctl read getattr write setattr append bind connect getopt setopt sh
}
type swiftapi_data_t;
type unconfined_t;
}

# 1. Define the Types
type swiftapi_t;
type swiftapi_exec_t;
type swiftapi_cfg_t;
type swiftapi_data_t;
type swiftapi_log_t;
type swiftapi_tmp_t;
type swiftapi_port_t, port_type;

# 2. Setup the Domain
domain_type(swiftapi_t)
domain_entry_file(swiftapi_t, swiftapi_exec_t)
init_daemon_domain(swiftapi_t, swiftapi_exec_t)

# 3. Fix the "Search" Denials (Directory Access)
# Allow systemd, the app, and the troubleshooter to enter the directory
allow { init_t swiftapi_t setroubleshootd_t } swiftapi_cfg_t:dir { read search open getattr };

[ Read 67 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_/ Go To Line M-E Redo

```

(2) File Context File

nano swiftapi.fc



```
May 2 7:56 AM
swift-terminal@fedora:~/selinux-policy/swiftapi — nano swiftapi.fc
~/selinux-policy/swiftapi

GNU nano 8.5 swiftapi.fc
# EXECUTABLES
/opt/swift-backend/venv/bin/uvicorn system_u:object_r:swiftapi_exec_t:s0
/opt/swift-backend/venv/bin/python.* system_u:object_r:swiftapi_exec_t:s0

# CONFIG
/opt/swift-backend(/.*)? system_u:object_r:swiftapi_cfg_t:s0

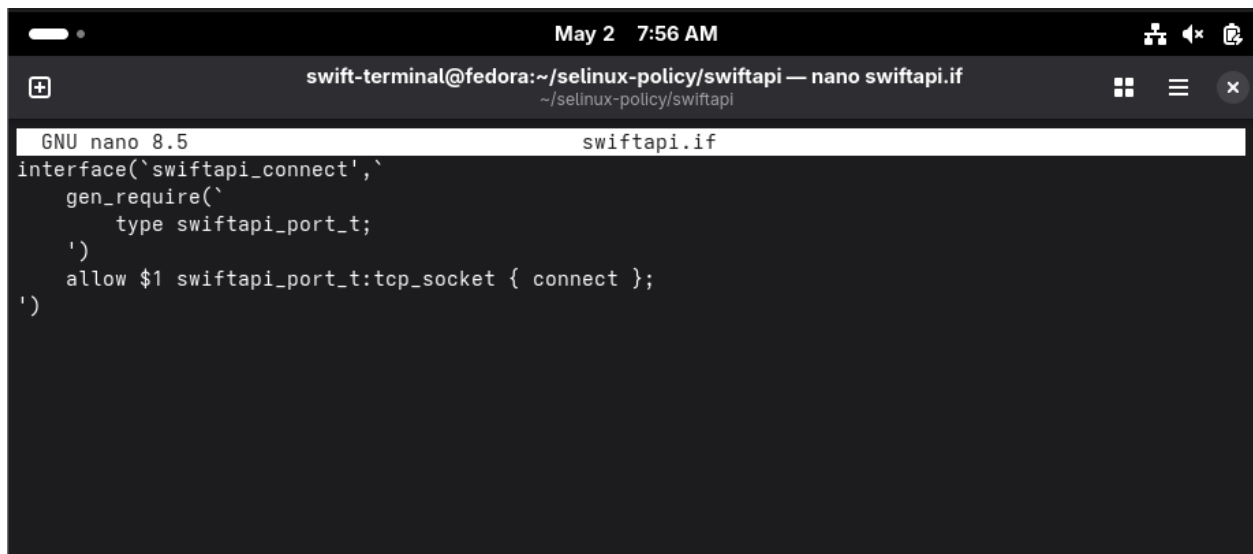
# DATA
/swift-lt(/.*)? system_u:object_r:swiftapi_data_t:s0

# LOGS
/var/log/swift-backend(/.*)? system_u:object_r:swiftapi_log_t:s0

# TEMP
/tmp/swiftapi(/.*)? system_u:object_r:swiftapi_tmp_t:s0
```

(3) Interface File

nano swiftapi.if



```
May 2 7:56 AM
swift-terminal@fedora:~/selinux-policy/swiftapi — nano swiftapi.if
~/selinux-policy/swiftapi

GNU nano 8.5 swiftapi.if
interface(`swiftapi_connect',`
    gen_require(`
        type swiftapi_port_t;
    ')
    allow $1 swiftapi_port_t:tcp_socket { connect };
')
```

5. Port Configuration (Critical Step)

By default, port 8000 is labeled http_port_t.

```
sudo semanage port -a -t swiftapi_port_t -p tcp 8000
```

Verify:

`sudo semanage port -l | grep swiftapi`

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ sudo semanage port -a -t swiftapi_port_t -p tcp 8000
Port tcp/8000 already defined, modifying instead
swift-terminal@fedora:~/selinux-policy/swiftapi$ sudo semanage port -l | grep swiftapi
swiftapi_port_t      tcp      8000
swift-terminal@fedora:~/selinux-policy/swiftapi$
```

Expected:

`swiftapi_port_t tcp 8000`

6. Compile and Load SELinux Module

6.1 Compile Policy

`make -f /usr/share/selinux/devel/Makefile`

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ make -f /usr/share/selinux/devel/Makefile
Compiling targeted swiftapi module
Creating targeted swiftapi.pp policy package
rm tmp/swiftapi.mod tmp/swiftapi.mod.fc
```

Install policy

`sudo semodule -i swiftapi.pp`

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ sudo semodule -i swiftapi.pp
```

6.4 Verify Installation

`sudo semodule -l | grep swiftapi`

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ sudo semodule -l | grep swiftapi
swiftapi
swift-terminal@fedora:~/selinux-policy/swiftapi$
```

7. Apply SELinux Labels

```
sudo restorecon -Rv /opt/swift-backend/  
sudo restorecon -Rv /swift-lt/  
sudo restorecon -Rv /var/log/swift-backend/
```

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ sudo restorecon -Rv /opt/swift-backend/  
sudo restorecon -Rv /swift-lt/  
sudo restorecon -Rv /var/log/swift-backend/  
restorecon: lstat(/var/log/swift-backend) failed: No such file or directory  
swift-terminal@fedora:~/selinux-policy/swiftapi$
```

7.1 Verify Labels

```
ls -ldZ /swift-lt  
ls -lZ /opt/swift-backend/.env  
ls -lZ /opt/swift-backend/venv/bin/uvicorn
```

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ ls -ldZ /swift-lt  
drwxr-x---. 1 swiftapi swift-ops system_u:object_r:swiftapi_data_t:s0 30 Mar 28 16:31 /swift-lt  
swift-terminal@fedora:~/selinux-policy/swiftapi$
```

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ ls -ldZ /swift-lt  
ls -lZ /opt/swift-backend/.env  
ls -lZ /opt/swift-backend/venv/bin/uvicorn  
  
drwxr-x---. 1 swiftapi swift-ops system_u:object_r:swiftapi_data_t:s0 30 Mar 28 16:31 /swift-lt  
total 0  
-rwxr-xr-x. 1 swiftapi swiftapi system_u:object_r:swiftapi_exec_t:s0 230 Mar 6 02:50 /opt/swift-back  
kend/venv/bin/uvicorn  
swift-terminal@fedora:~/selinux-policy/swiftapi$
```

8. Testing and Verification

8.1 Service Domain Check

```
ps -eZ | grep uvicorn
```

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ ps -eZ | grep uvicorn  
system_u:system_r:swiftapi_t:s0      8973 ?          00:00:00 uvicorn
```

Expected:

```
system_u:system_r:swiftapi_t:s0
```

8.2 API Test

```
curl http://127.0.0.1:8000/health
```



```
system_u:system_r:swiftapi_t:s0 8573 ? 00:00:00 swift-terminal@fedora:~/selinux-policy/swiftapi$ curl http://127.0.0.1:8000/health
{"status":"ok"}swift-terminal@fedora:~/selinux-policy/swiftapi$
```

Expected:

```
{"status":"ok"}
```

8.3 Unauthorized File Access Test

`sudo -u audit1 ls /swift-lt/LT1`

```
{"status":"ok"}swift-terminal@fedora:~/selinux-policy/swiftapi$ sudo -u audit1 ls /swift-lt/LT1
ls: cannot access '/swift-lt/LT1': Permission denied
swift-terminal@fedora:~/selinux-policy/swiftapi$
```

Expected:

Permission denied

Check logs:

`sudo ausearch -m AVC -ts recent`

```
time->Sat May  2 07:29:54 2026
type=PROCTITLE msg=audit(1777688994.016:2668): proctitle=726573746F72655636F6E002D527646002F73776966742D6C742F
type=PATH msg=audit(1777688994.016:2668): item=0 name="/swift-lt" inode=211297 dev=00:20 mode=040750 ouid=972 ogid=1007 rdev=00:00 obj=system_u:object_r:swiftapi_data_t:s0 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1777688994.016:2668): cwd="/home/swift-terminal/selinux-policy/swiftapi"
type=SYSCALL msg=audit(1777688994.016:2668): arch=c000003e syscall=257 success=no exit=-13 a0=ffffff9c a1=55c39cc9e9e0 a2=90800 a3=0 items=1 ppid=9114 pid=9115 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=3 comm="restorecon" exe="/usr/bin/setfiles" subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key="swift_files"
type=AVC msg=audit(1777688994.016:2668): avc: denied { read } for pid=9115 comm="restorecon" name="/swift-lt" dev="sda3" ino=211297 scontext=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 tcontext=system_u:object_r:swiftapi_data_t:s0 tclass=dir permissive=0
swift-terminal@fedora:~/selinux-policy/swiftapi$
```

8.5 Config Protection Test

`sudo -u operator1 echo "test" >> /opt/swift-backend/.env`

```
swift-terminal@fedora:~/selinux-policy/swiftapi$ sudo -u operator1 echo "test" >> /opt/swift-backend/.env/text.txt
bash: /opt/swift-backend/.env/text.txt: Permission denied
```

Expected:

Permission denied

Conclusion

A **strict SELinux policy** was successfully implemented on **Fedora Linux** for the swiftapi backend.

Key achievements:

- Replaced permissive domains with **custom confined domain (swiftapi_t)**
- Enforced **least privilege principle**
- Protected:
 - Financial data (/swift-lt)
 - Configuration files
 - Network port 8000
- Prevented:
 - Unauthorized file access
 - Privilege escalation
 - Port misuse

The system now enforces **Mandatory Access Control (MAC)** at kernel level, ensuring that even root-level processes cannot violate policy rules without explicit permission.

This implementation significantly strengthens the **defense-in-depth architecture** of the SWIFT Logical Terminal Infrastructure.